

Date: 13/10/2025

Submission Date: 20/10/2025 (11:55 PM)

Note: Below is the list of questions on **MOS Transistors** and **Small-Signal Parameters of a MOS Transistor**. You are required to:

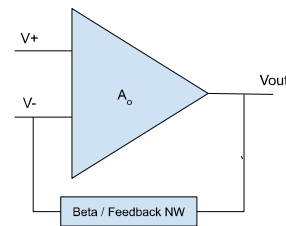
- Perform the simulations using **Cadence Virtuoso** with **SCL 180 nm PDK**.
- Prepare a detailed report based on your work.
- Submit the report for evaluation.

Report Guidelines:

- Include a **Conclusion** section at the end, briefly summarizing your insights and observations from the design process.
 - Name the report file as: RollNumber_Assignment6.pdf
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1. Design an Operational **Transconductance Amplifier (OTA)** for the following specifications.
Form Assignment 5

- Differential Gain ≥ 40 , CMRR $\geq 70\text{dB}$
- ICMR: $0.70\text{ V} - 1.4\text{ V}$
- Output Swing $> 800\text{m V}$, Power $\leq 20\text{ uW}$
- Load Capacitance = 1pF , GBW $\geq 10\text{MHz}$



- a. Use the negative feedback concept and control the gain with the help of the beta factor (Choose the R1 and R2 based on the loading effect).
 - Case1 : If β is $1/100$, then calculate the Gain, GBW, BW, phase, Power, and error from calculated and from simulated values (only for gain). Your error should be less than 10%.
 - Case2 : Repeat case 1, If β is $1/200$.
 - b. If your error is not less than 10% then redesign the OTA for the higher gain, and then check your specification and resimulate both cases. Make a comparison table for both cases.
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Report should include,

- Transistor level circuit diagram (full system and individual sub-blocks) and sizing (with overdrive voltage, gm, gds and region) table of all transistors.
- Step-by-step design methodology/derivations/calculations (in brief).
- Overall power consumption and its block-wise breakup should be provided.
- Plots should be neatly labelled, and labels should be clearly visible
- Final results should be provided in tabular format
- Any other information as deemed necessary